

CUSTOMER SEGMENTATION USING RFM ANALYSIS

(A Power BI Based Customer Analytics Project)

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Tools Used:

- Microsoft Power BI
- Microsoft Excel

Project Type:

- Data Analytics
- Business Intelligence
- Customer Analytics

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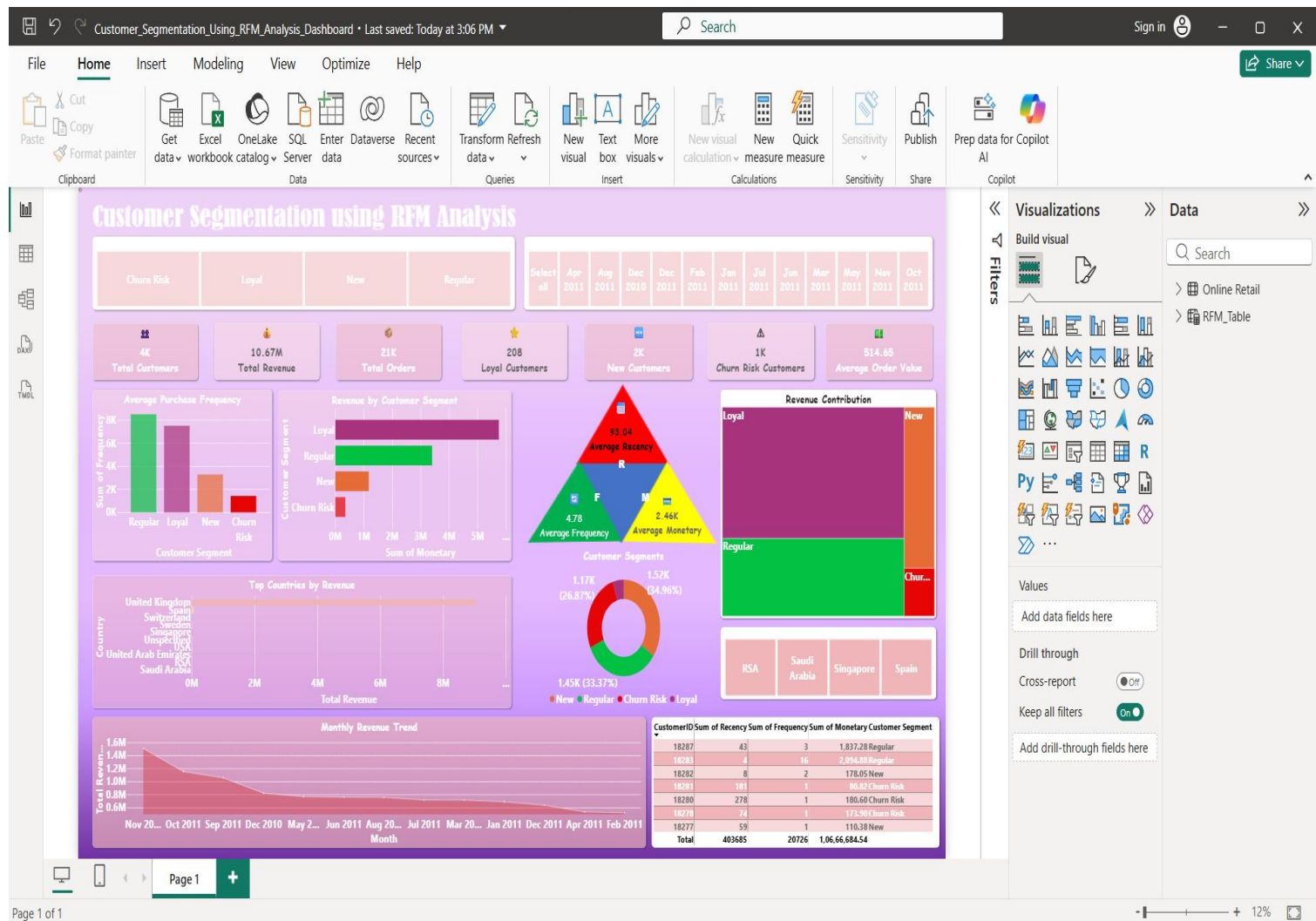


Figure 1: Interactive Customer Segmentation Dashboard created using Microsoft Power BI.

Introduction

This project focuses on **Customer Segmentation using RFM (Recency, Frequency, Monetary) Analysis** in Microsoft Power BI. The primary objective is to analyze customer purchasing behavior and classify customers into meaningful segments based on their purchasing patterns.

The dashboard transforms raw transactional retail data into actionable business insights through data cleaning, RFM metric calculation, customer segmentation, KPI development, and interactive visualizations.

The dashboard enables businesses to identify loyal customers, detect customers at risk of churning, recognize new customers, and develop targeted marketing strategies to improve customer retention and increase revenue.

Project Objectives

The primary objectives of this project are:

- Import and clean the transactional retail dataset.
- Prepare customer-level data for RFM analysis.
- Calculate Recency, Frequency, and Monetary metrics.
- Segment customers into meaningful groups.
- Analyze customer purchasing behavior.
- Create Key Performance Indicators (KPIs).
- Build an interactive dashboard using Microsoft Power BI.
- Generate business insights and marketing recommendations.

Dataset Description

The dataset used for this project is the **Online Retail Dataset**, containing customer transaction records from an online retail store.

Key Columns Used

- InvoiceNo
- StockCode
- Description
- Quantity
- InvoiceDate
- UnitPrice
- CustomerID
- Country
- TotalAmount (Created)
- Frequency (Calculated)
- Monetary (Calculated)
- Recency (Calculated)
- Customer Segment (Calculated)

These fields were used to calculate customer purchasing behavior and build customer segments using RFM analysis.

Dataset Summary

Before analysis, the dataset underwent several data quality checks to ensure accurate calculations and reliable business insights.

Attribute	Description
Dataset	Online Retail
Records	Transaction-level sales data
Time Period	December 2010 – December 2011
Customers	Multiple Customers
Countries	Multiple Countries
Measures	Quantity, Unit Price, Total Amount

Data Cleaning Process

Before performing RFM analysis, the dataset was cleaned to improve data quality and eliminate inconsistencies.

Data Cleaning Steps Performed

- Imported the dataset into Microsoft Power BI.
- Removed blank Customer IDs.
- Removed returned orders ($\text{Quantity} \leq 0$).
- Removed records with $\text{Unit Price} \leq 0$.
- Verified data types for all columns.
- Converted InvoiceDate into Date format.
- Created TotalAmount column using $\text{Quantity} \times \text{UnitPrice}$.
- Verified numerical columns for accurate calculations.

These preprocessing steps ensured that the dataset was suitable for customer behavior analysis.

RFM Analysis

RFM Analysis is a customer segmentation technique that evaluates customer value using three important metrics.

Recency (R)

Measures how recently a customer made their last purchase.

Lower Recency values indicate customers who purchased recently and are more likely to remain active.

Frequency (F)

Measures how often a customer places orders.

Higher Frequency indicates stronger customer engagement.

Monetary (M)

Measures the total amount spent by a customer.

Higher Monetary values indicate customers who contribute more revenue.

Customer Segmentation

Customers were segmented into the following groups based on their RFM scores:

Loyal Customers

- Purchase frequently
- Spend more money
- Purchased recently

These customers are the most valuable customers.

Regular Customers

- Moderate purchase frequency
- Moderate spending
- Potential to become loyal customers

New Customers

- Recently acquired customers
- Low purchase frequency
- Require engagement campaigns

Churn Risk Customers

- Long time since last purchase
- Low purchase frequency
- High probability of leaving

Dashboard Analysis

The dashboard provides a comprehensive overview of customer purchasing behavior.

Customer Segment Distribution

A Donut Chart was created to visualize the proportion of customers in each customer segment.

This enables quick identification of customer composition.

Revenue by Customer Segment

A Clustered Bar Chart compares total revenue generated by different customer segments.

This identifies which customer group contributes the highest revenue.

Average Purchase Frequency

A Column Chart compares purchase frequency among customer segments.

It highlights customer engagement levels.

Monthly Revenue Trend

A Line Chart analyzes revenue generated across different months.

This helps identify seasonal purchasing trends.

Top Countries by Revenue

A Bar Chart compares revenue generated across countries.

This identifies the highest contributing regions.

Revenue Contribution

A Treemap visualizes revenue contribution by customer segment.

This provides an intuitive understanding of revenue distribution.

Customer RFM Summary

A Table visual displays:

- Customer ID

- Recency
- Frequency
- Monetary
- Customer Segment

This provides a detailed customer-level analysis.

Key Performance Indicators (KPIs)

The dashboard includes the following KPIs:

Total Customers

Represents the total number of unique customers.

Total Revenue

Represents the total sales generated.

Total Orders

Represents the total number of unique invoices.

Average Recency

Represents the average number of days since customers made their last purchase.

Average Frequency

Represents the average number of purchases made by customers.

Average Monetary

Represents the average amount spent by customers.

Loyal Customers

Represents the number of customers classified as Loyal.

New Customers

Represents the number of customers classified as New.

Churn Risk Customers

Represents the number of customers likely to stop purchasing.

Average Order Value

Represents the average revenue generated per order.

Interactive Features

The dashboard contains interactive slicers that allow users to filter data dynamically.

Available Filters

- Country
- Customer Segment
- Month

These filters allow users to explore customer behavior from different perspectives.

Key Findings

The following insights were obtained from the analysis:

- Loyal customers contribute the highest share of revenue.
- Churn Risk customers have not purchased recently and require retention efforts.
- New customers have recently joined but have lower purchase frequency.
- Regular customers have the potential to become loyal customers.
- A small percentage of customers contributes a significant portion of total sales.
- Customer purchasing behavior varies across different countries.

Tools Used

- Microsoft Power BI
- Microsoft Excel

Project Type

- Data Analytics
- Business Intelligence Dashboard
- Customer Analytics
- Customer Segmentation
- RFM Analysis

Skills Demonstrated

- Data Cleaning
- Data Transformation
- Data Modeling
- DAX Calculations
- RFM Analysis
- Customer Segmentation
- KPI Development
- Dashboard Design
- Business Intelligence Reporting
- Data Visualization using Power BI

Marketing Recommendations

Loyal Customers

- Introduce VIP membership programs.
- Offer exclusive discounts.
- Provide early access to new products.
- Reward repeat purchases.

Regular Customers

- Recommend related products.
- Offer bundle discounts.
- Provide cashback offers.

New Customers

- Send welcome emails.
- Offer first-purchase discounts.
- Recommend popular products.

Churn Risk Customers

- Launch win-back campaigns.
- Offer personalized discounts.
- Send reminder emails.
- Re-engage through targeted promotions.

Future Scope

Future improvements may include:

- Predictive customer churn analysis using Machine Learning.
- Customer Lifetime Value (CLV) prediction.
- Personalized product recommendation systems.
- Automated customer segmentation.
- Real-time dashboard integration using live retail databases.

Conclusion

The **Customer Segmentation using RFM Analysis Dashboard** successfully transforms raw retail transaction data into meaningful customer insights. Through data cleaning, RFM metric calculation, customer segmentation, KPI development, and interactive visualizations, the dashboard provides a comprehensive understanding of customer purchasing behavior.

This project demonstrates practical skills in **Data Analytics, Business Intelligence, Customer Segmentation, DAX, Data Modeling, and Power BI Dashboard Development**. The insights and recommendations generated from this analysis can support customer retention strategies, improve marketing effectiveness, and enable data-driven business decision-making.